CIRCUIT WALKTHROUGH

4-Channel Electronic Load

60 V · 20 A · 300 W

Constant-current sink · 20-bit measurement · hardware fail-safe · WiFi control

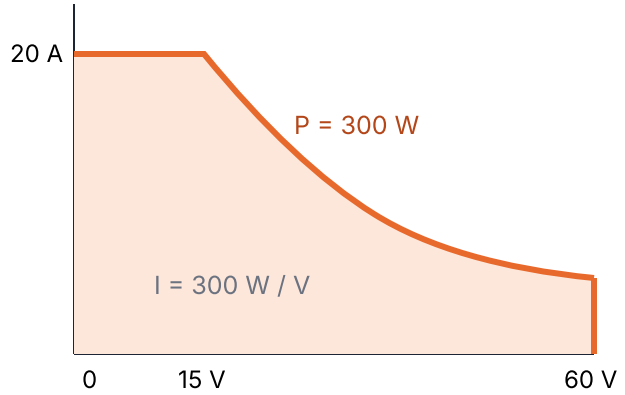
rev 0.3 · IV3HGA · October 2026

OVERVIEW

What the load does

PARAMETER	VALUE
Input voltage	0 – 60 V (hardware OVP trips at 64.7 V)
Current	4 channels × 0 – 5 A = 20 A
Power	4 × 75 W = 300 W continuous
Mode	Constant current per channel; CP / CR in firmware
Setpoint	16-bit DAC → 76 µA per LSB per channel
Measurement	20-bit INA228 per channel, Kelvin shunt, remote sense
Control	WiFi — ESP8266 (Wemos LOLIN D1 mini Pro)
Protection	OVP, OTP, over-current ALERT, reverse polarity, fuse

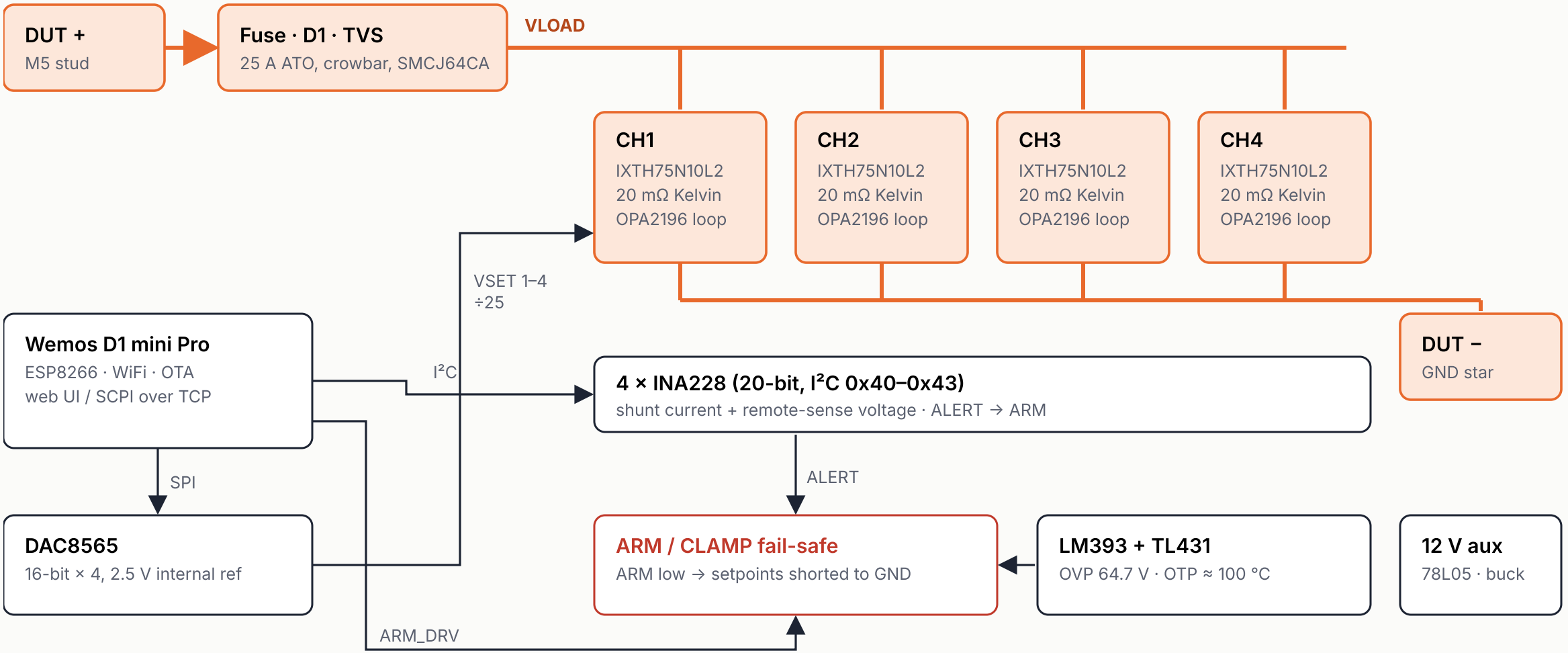
Safe operating envelope



Below 15 V the limit is current (20 A); above 15 V it is power. Firmware enforces it, INA228 ALERT backs it up in hardware.

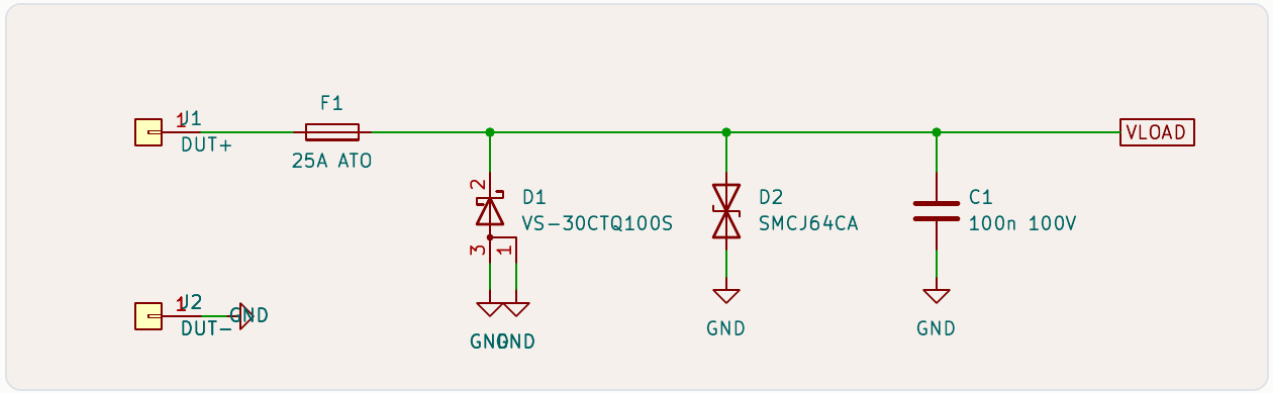
ARCHITECTURE

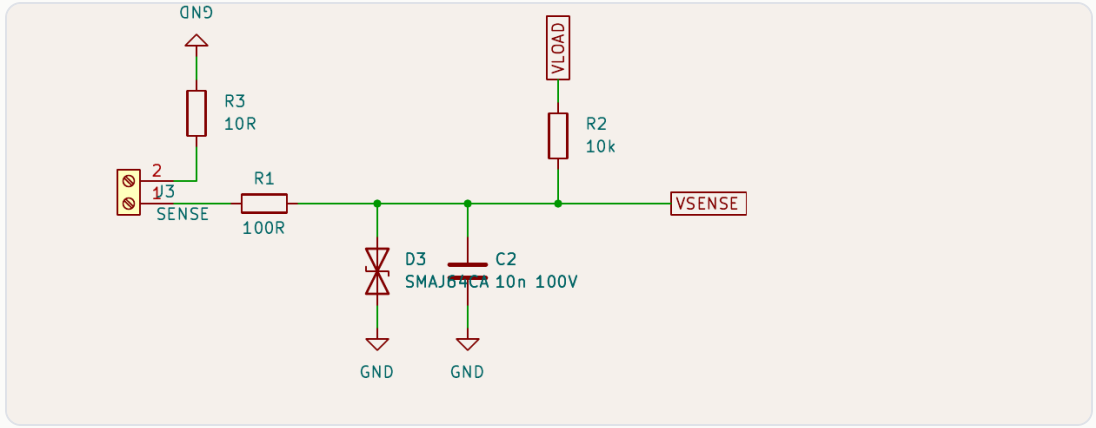
Block diagram



Orange = power path (up to 20 A). Black = signal / control. The whole controller is referenced to DUT−; the 12 V adaptor must be isolated.

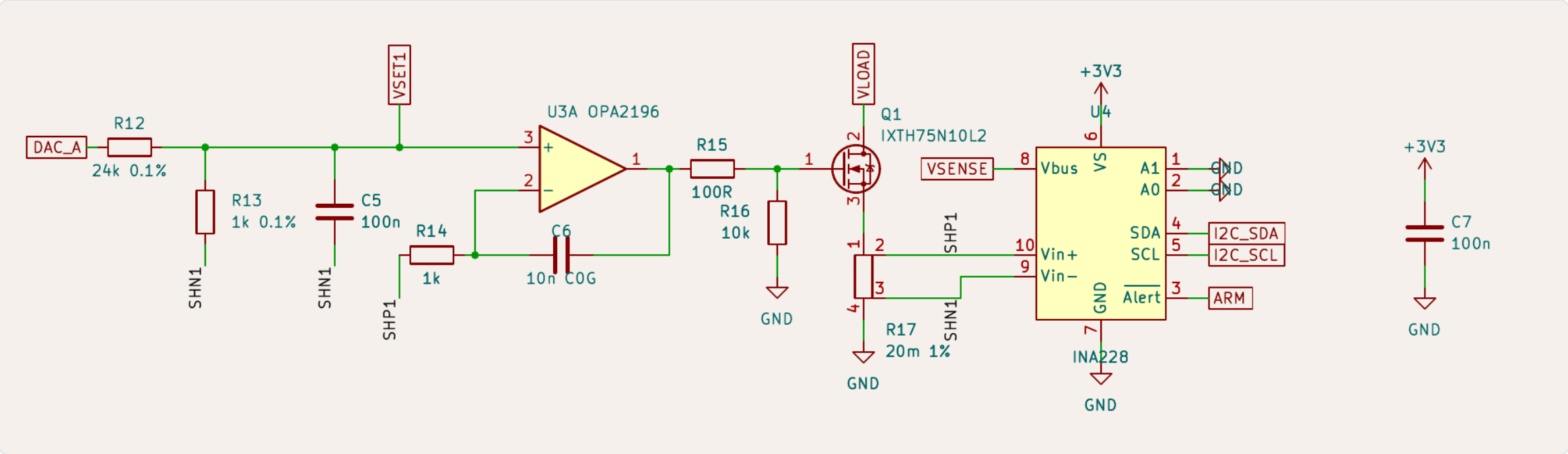
Input stage and remote sense





- **F1 — 25 A ATO blade fuse** in series with DUT+. Cheap, replaceable, rated for the 20 A full load with margin.
- **D1 — VS-30CTQ100S** (D²PAK Schottky) is a reverse-polarity crowbar: if the DUT is connected backwards it conducts and blows F1 before anything else sees negative voltage.
- **D2 — SMCJ64CA** clamps transients on VLOAD; C1 keeps the bus quiet at HF.
- **Remote sense (J3)** measures voltage at the DUT terminals, not at the studs. R1 + D3 protect the INA228 inputs.
- **R2 10 k** pulls VSENSE to VLOAD when the sense leads are not connected, so readings stay sensible. R3 10 Ω ties SENSE− to ground.

The constant-current loop



- **R12 / R13 (24 k / 1 k, 0.1 %)** divide the DAC output by 25. The bottom of the divider is the shunt's *cold* Kelvin pad (SHN), so ground drops do not enter the setpoint.
- **OPA2196** drives the gate until SHP − SHN equals the setpoint. R14 + C6 (1 nF C0G, rev 0.2) set the integrator; R15 100 Ω is the gate stopper, R16 keeps the gate off when unpowered.

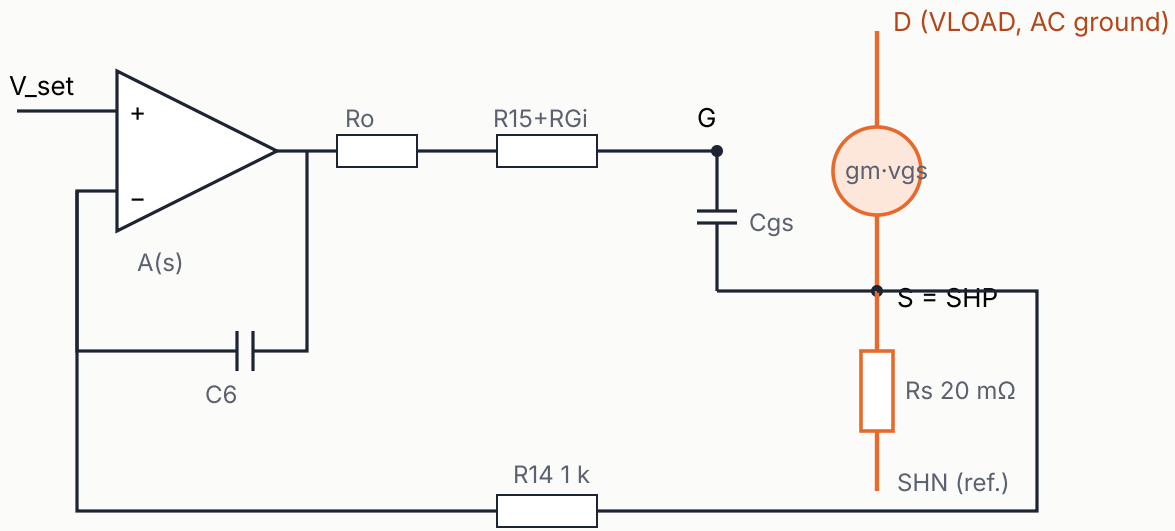
$$V_{set} = V_{DAC} / 25$$

$$I = V_{set} / R_{sh} = V_{DAC} / (25 \times 20\ m\Omega)$$

$$I = 2\ A/V \rightarrow 5\ A \text{ at } 2.5\ V$$

$$1\ LSB = 2.5\ V / 65536 \rightarrow 76\ \mu A$$

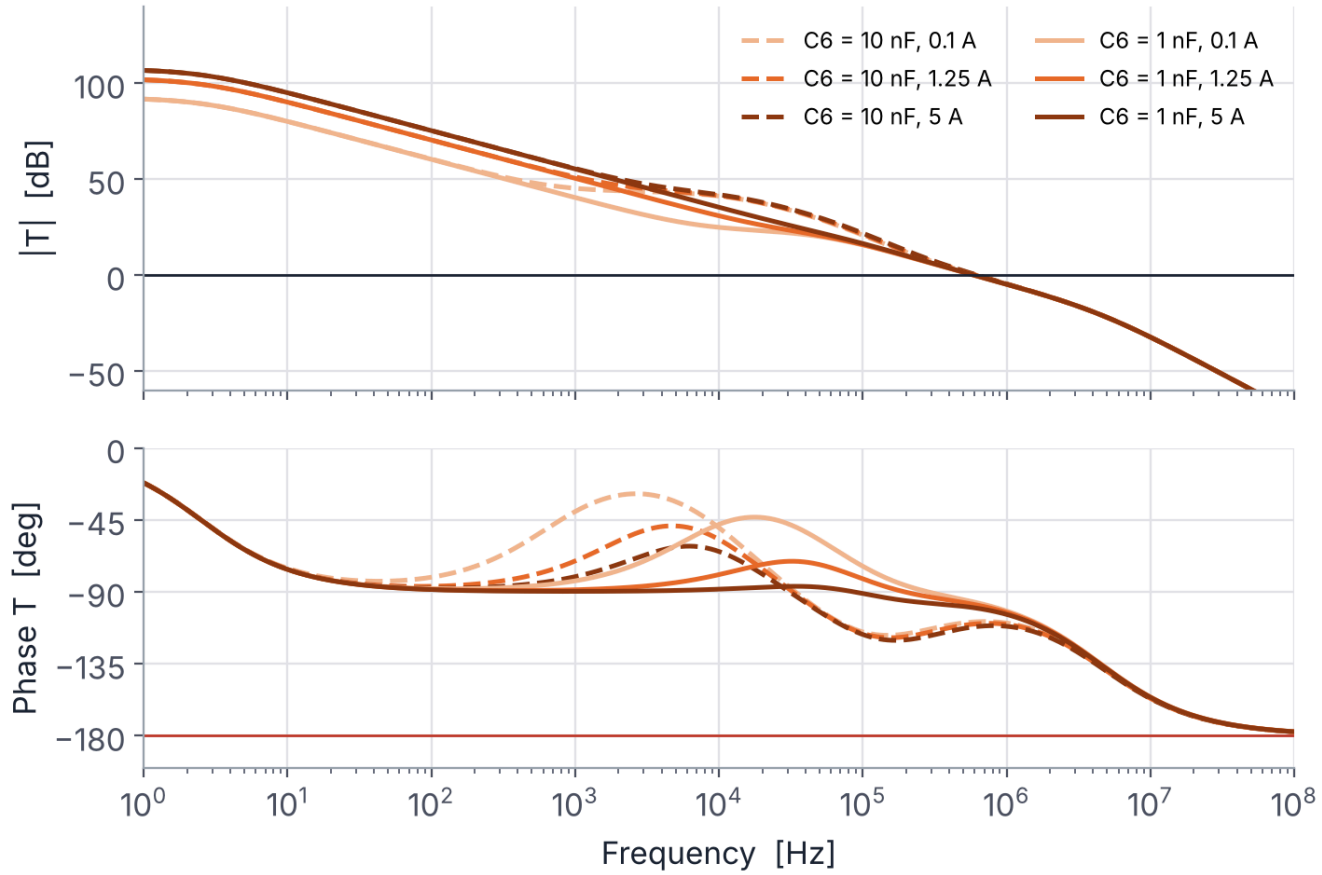
Small-signal model of one channel



PARAMETER	VALUE	SOURCE
OPA2196 GBW / A0 / 2nd pole	2.5 MHz / 120 dB / 4.5 MHz	datasheet / assumed
Open-loop Ro	300 Ω (worst 600 Ω)	assumed
Ciss / Crss, RGi	8.1 nF / 0.35 nF, 3 Ω	IXTH75N10L2 ds
gm	44 S at 37.5 A, square law	ds (gfs typ)

- **Plant.** The MOSFET is a source follower degenerated by the shunt: $V_{fb}/V_g = k = g_m R_s/(1+g_m R_s)$, with a gate pole set by (Ro + R15 + RGi)·Ciss.
- **k depends on current:** g_m = 2.3 S at 0.1 A → k = 0.04; 8 S at 1.25 A → 0.14; 16 S at 5 A → 0.24. Loop gain changes 6× across the range.
- **Controller.** R14 + C6 make an integrator, so the DC error is zero. Outer-loop crossover ≈ k / (2π R14 C6).
- **Setpoint filter.** R12‖R13 = 960 Ω with C5 gives a pole at 1/(2π·960·C5).
- The loop is broken at the op-amp's internal source: the return ratio includes both feedback paths (through the MOSFET, and directly through C6).

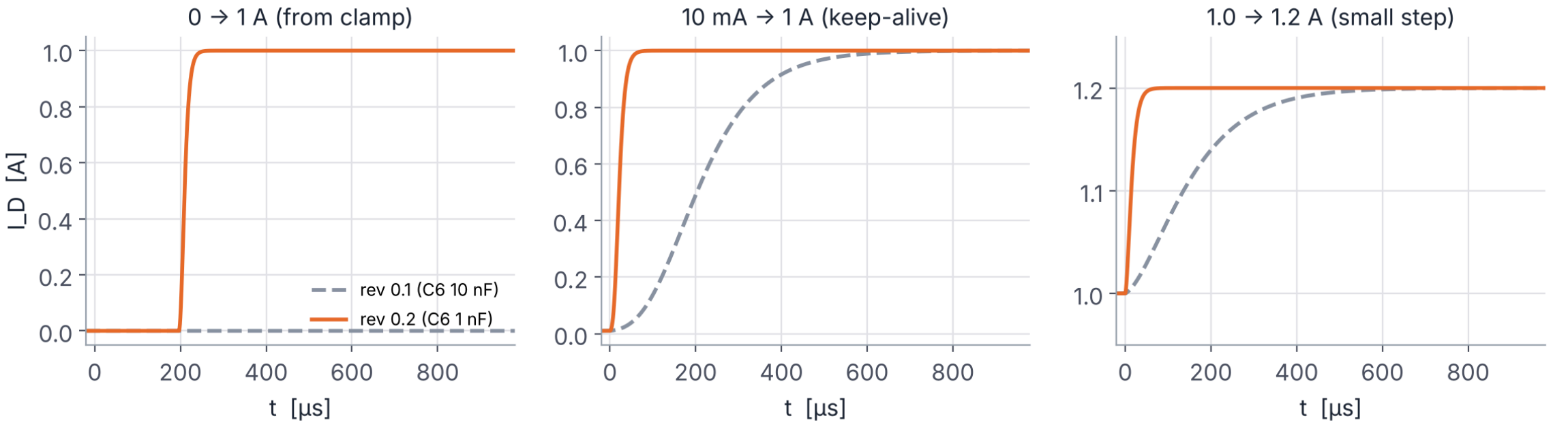
Loop gain: rev 0.1 vs rev 0.2



PHASE MARGIN	0.1 A	1.25 A	5 A
rev 0.1 · C6 = 10 nF	71°	70°	68°
rev 0.2 · C6 = 1 nF	84°	82°	80°

- **Stable in all cases.** The total loop crosses 0 dB near 600 kHz, where the direct C6 path dominates; the gain margin is never reached.
- **Robust to model errors:** Ro 600 Ω or Ciss +50 % keep PM above 64° (10 nF) and 80° (1 nF).
- **Sensitive parts:** R15 down to 47 Ω drops PM to 49°. A resistor in series with C6 (PI zero) drops it to 30° or less — do not add one.

Closed-loop response: why rev 0.2 changed C5 and C6



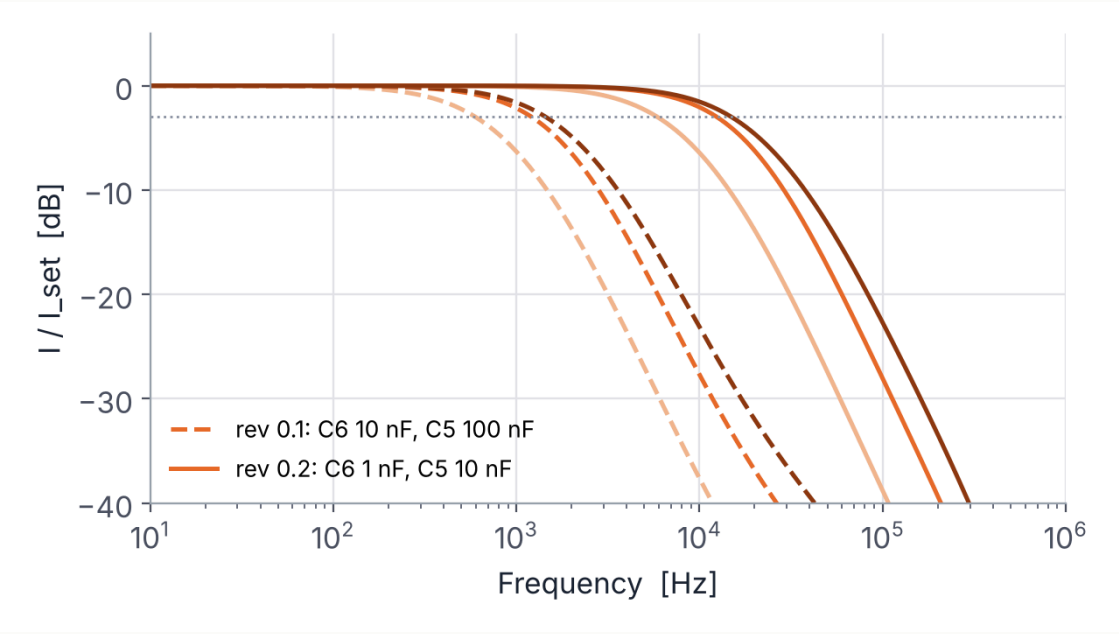
- **Small signal** (1.0 → 1.2 A): 10–90 % in ~290 µs in rev 0.1, ~28 µs in rev 0.2. No overshoot in either case.
- **Bandwidth** I/I_{set}: 0.6–1.5 kHz in rev 0.1 (limited by both the integrator and the 1.66 kHz setpoint filter), 6–15 kHz in rev 0.2.
- **Turn-on from 0 A** is a large-signal dead time: the op-amp output sits at 0 V and must ramp the gate to V_{th} at a rate $V_{set}/(R14 \cdot C6)$.

$$t_d \approx V_{th} \cdot R14 \cdot C6 / V_{set}$$

1 A, 3.5 V: 1.75 ms (10 nF) → 175 µs (1 nF)

Compensation adopted in rev 0.2

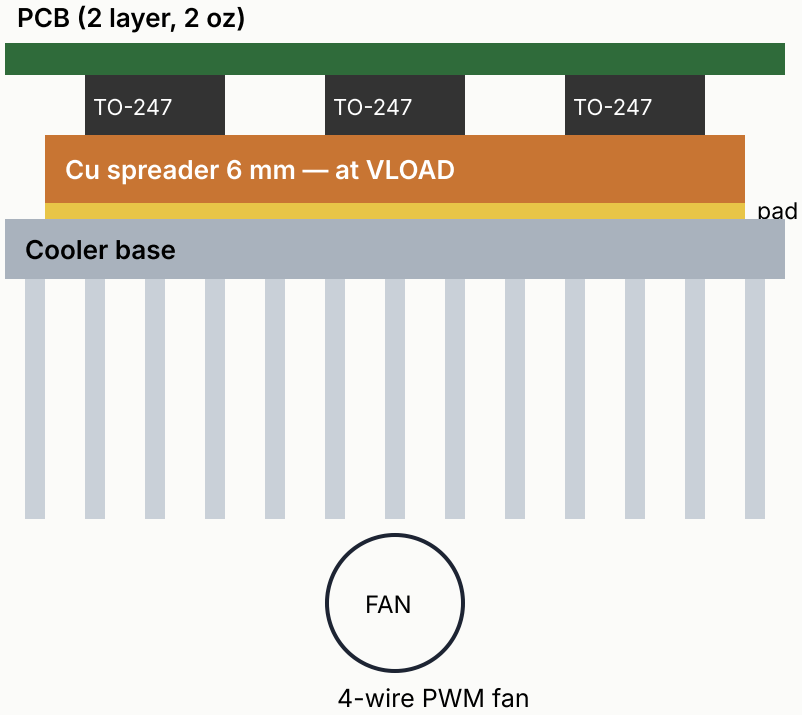
PART	REV 0.1	REV 0.2	EFFECT
C6, C9, C12, C15	10 nF C0G	**1 nF C0G**	10× loop bandwidth, PM 70° → 80°
C5, C8, C11, C14	100 nF	**10 nF C0G**	setpoint pole 1.66 → 16.6 kHz
R15 (gate)	100 Ω	keep ≥ 100 Ω	isolates Ciss from the op-amp
Firmware	—	**10 mA keep-alive** when armed	turn-on 200 µs → 39 µs



- Only capacitor values change: same footprints, no re-layout.
- The keep-alive holds the gate just above threshold, so every step starts in the linear region. 10 mA is 0.2 % of full scale and the INA228 measures it, so firmware can subtract it.
- Bandwidth still varies with current ($k \propto g_m$): slowest at low current. That is acceptable for a DC load; for pulsed tests use currents ≥ 0.5 A.
- **Bench check:** square-wave the setpoint between 1.0 and 1.2 A into a stiff supply, then repeat with 1 m of DUT cable; look for ringing on the shunt voltage.
- Model assumptions: OPA2196 2nd pole and Ro estimated; DUT treated as an ideal voltage source.

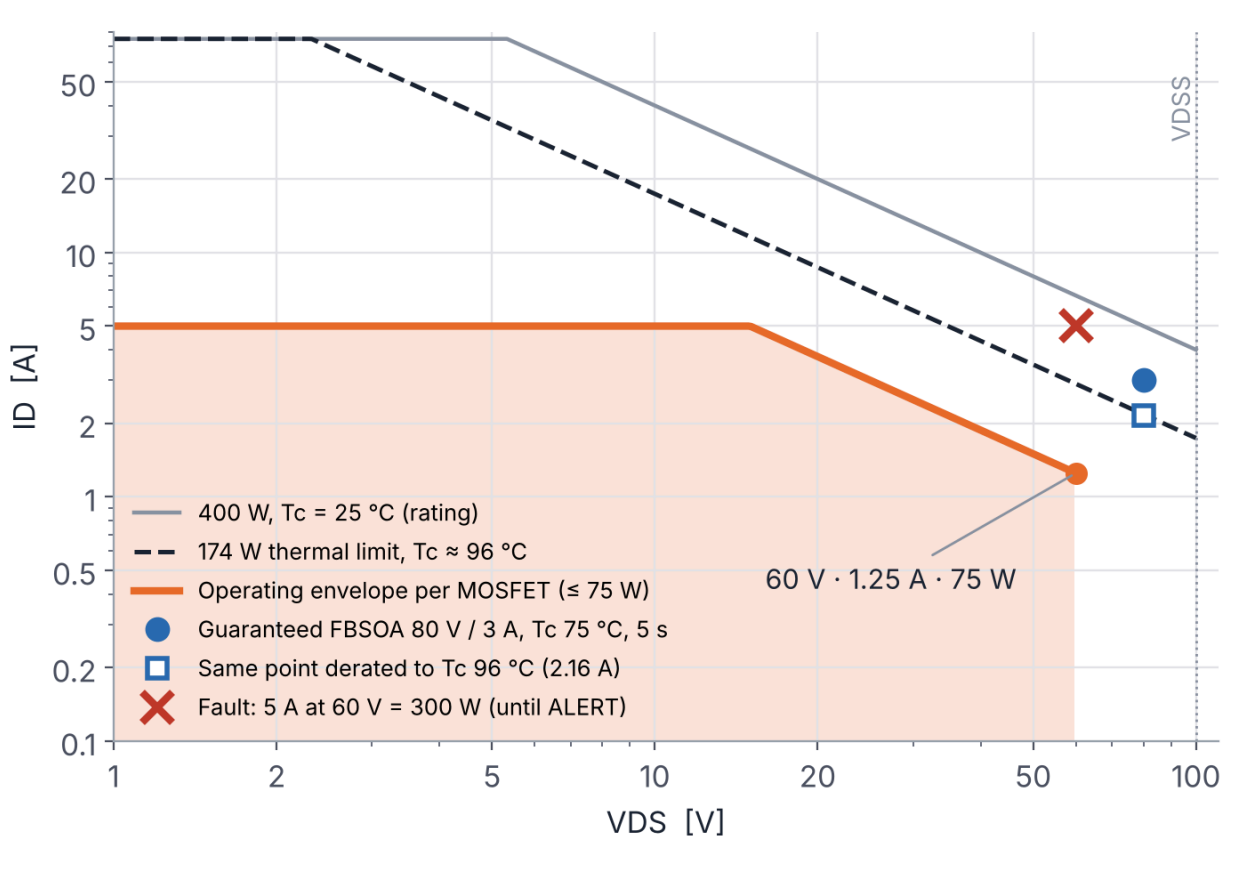
MOSFETs, shunts and heat

- **IXTH75N10L2 (Linear L2)** — 100 V, TO-247. Built for linear mode: its forward-bias SOA stays wide where normal switching MOSFETs suffer thermal runaway (hot spots). Each one dissipates at most 75 W.
- **WSK2512 20 mΩ, 4-terminal** shunt: 0.5 W at 5 A, Kelvin pads routed straight to the INA228 and to the loop.
- **Mounting:** the four TO-247 sit *under* the PCB, screwed onto a copper spreader (80 × 70 × 6 mm). The tabs are drains = VLOAD, so the spreader is live.
- **Spreader → cooler** through an insulating thermal pad rated ≥ 100 V. Cooler: SP3/TR4 class with a 4-wire PWM fan.
- **Two NTC 10 k** glued to the spreader: NTC1 feeds the hardware OTP comparator and the ESP8266 ADC.



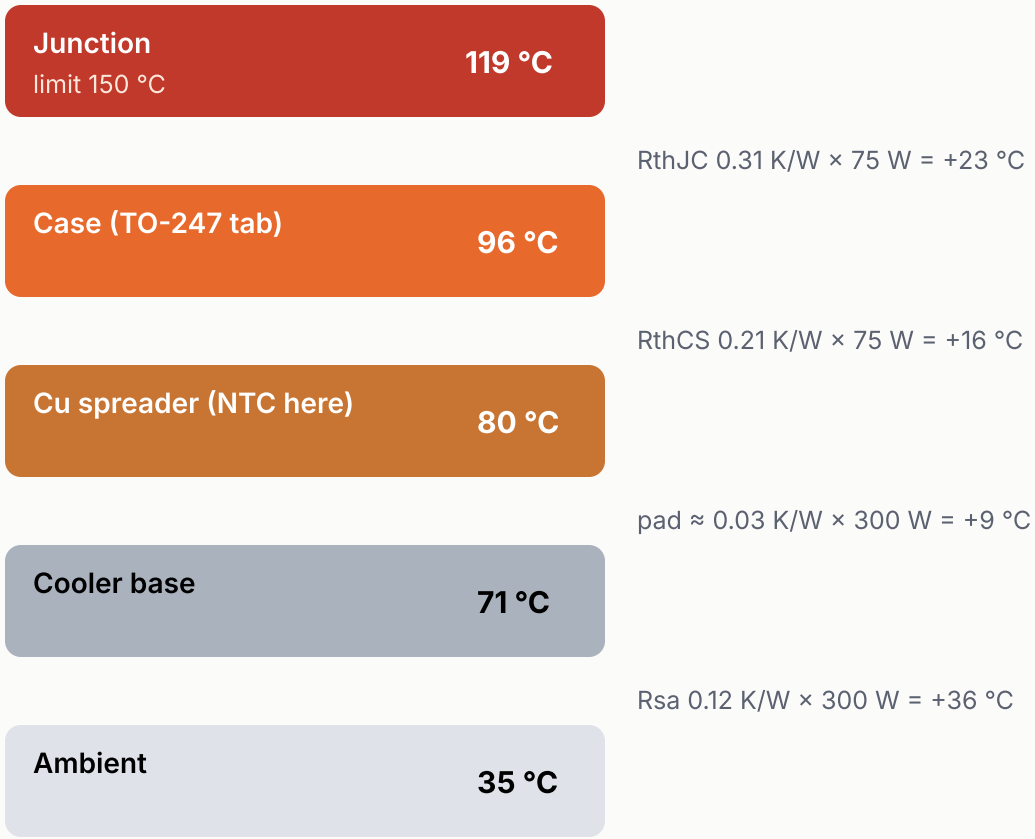
Thermal budget: 300 W total. The spreader evens out the four sources before heat crosses the insulating pad.

Operating point vs forward-bias SOA



- **Worst steady point per MOSFET:** 60 V × 1.25 A = 75 W (firmware limit I ≤ 75 W / V).
- **Guaranteed FBSOA** (datasheet): 80 V, 3 A = 240 W for 5 s at T_C = 75 °C — tested at a higher voltage than ours.
- **Derated to our case temperature** (≈ 96 °C, next slide):
  `240 W × (150 − 96)/(150 − 75) ≈ 173 W`
  → **2.3× margin** over 75 W. The pure thermal limit gives the same: (150 − 96)/0.31 = 174 W.
- **Fault case:** a wrong setpoint at 60 V could ask 5 A = 300 W. Set the INA228 power ALERT at ~80 W with 280 µs conversion and no averaging → trip in < 1 ms (≈ 0.3 J). Check against the 1 ms line of datasheet Fig. 14.
- Lines are computed from datasheet ratings, not digitised from the SOA figures.

Junction temperature budget at full load



- Per-device terms use 75 W; spreader, pad and cooler carry all 300 W.
- **Margin:** T_J ≈ 119 °C at 35 °C ambient, 31 °C below the 150 °C limit.
- **OTP consistency:** the comparator trips at ~100 °C on the spreader, so even at the trip point T_J ≈ 100 + 39 = 139 °C < 150 °C.
- **Cooler requirement** to run 300 W continuously without tripping OTP:

  ```
  Rsa + Rpad ≤ (100 − 35) / 300 = 0.22 K/W
  ```

- Assumed: 35 °C ambient; cooler 0.12 K/W at full fan (measure it); pad 80 × 70 mm, 0.5 mm, k ≈ 3 W/m·K; thermal grease under each TO-247 (datasheet RthCS).

MEASUREMENT

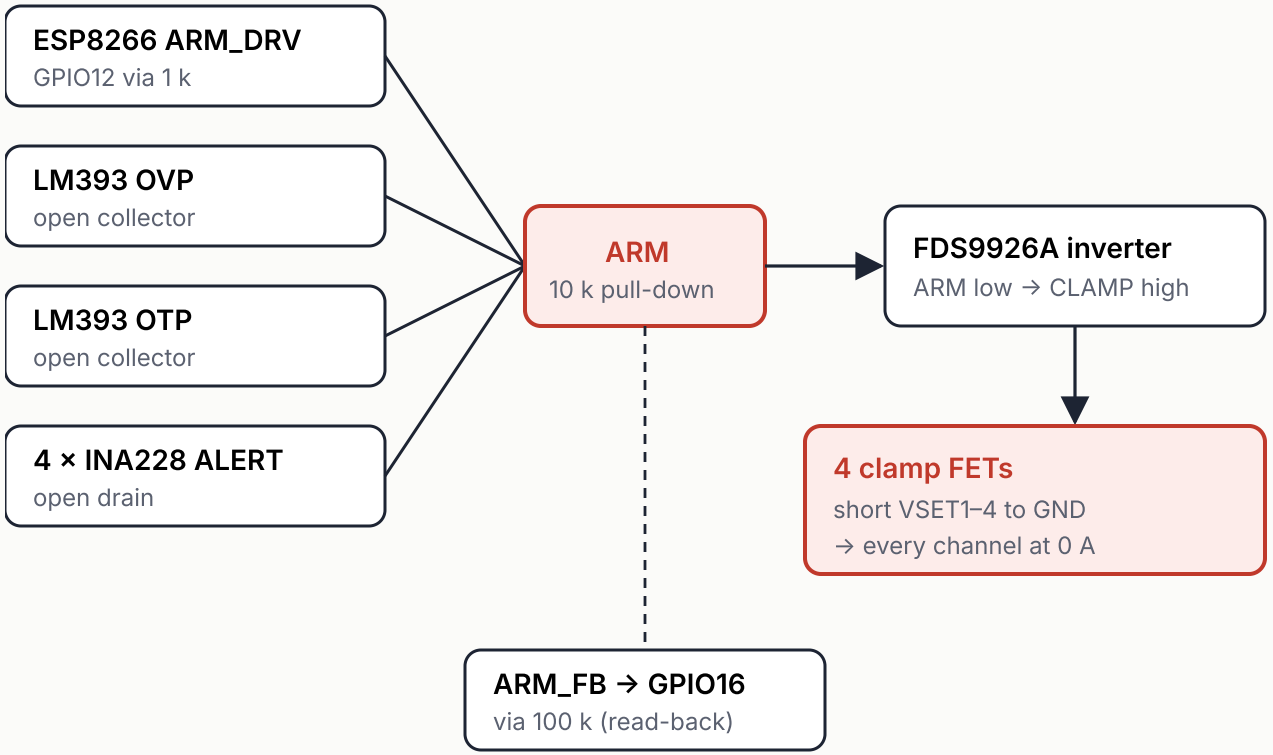
INA228: one precision monitor per channel

- **20-bit ΔΣ ADC**, 85 V common-mode: measures the shunt voltage (current) and VSENSE (voltage at the DUT) directly, no level shifting.
- Computes **power, energy and charge** internally — useful for battery capacity tests without MCU integration.
- **I²C addresses** set by A0: GND / VS / SDA / SCL → 0x40, 0x41, 0x42, 0x43. One shared bus with 2.2 k pull-ups (it also reaches the front panel).
- **ALERT** (open-drain) is wired to ARM: when a programmed limit (over-current, over-power, over-voltage) is exceeded, the INA228 shuts the load down *in hardware*, without waiting for firmware.
- Kelvin traces from the shunt run as a pair straight to pins 9–10.

CHANNEL	INA228	A0	ADDR
CH1	U4	GND	`0x40`
CH2	U5	VS	`0x41`
CH3	U7	SDA	`0x42`
CH4	U8	SCL	`0x43`

Resolution check: 5 A × 20 mΩ = 100 mV, so the ±163.84 mV range is used. LSB 312.5 nV → about 16 µA on a 20 mΩ shunt, well below the 76 µA DAC step.

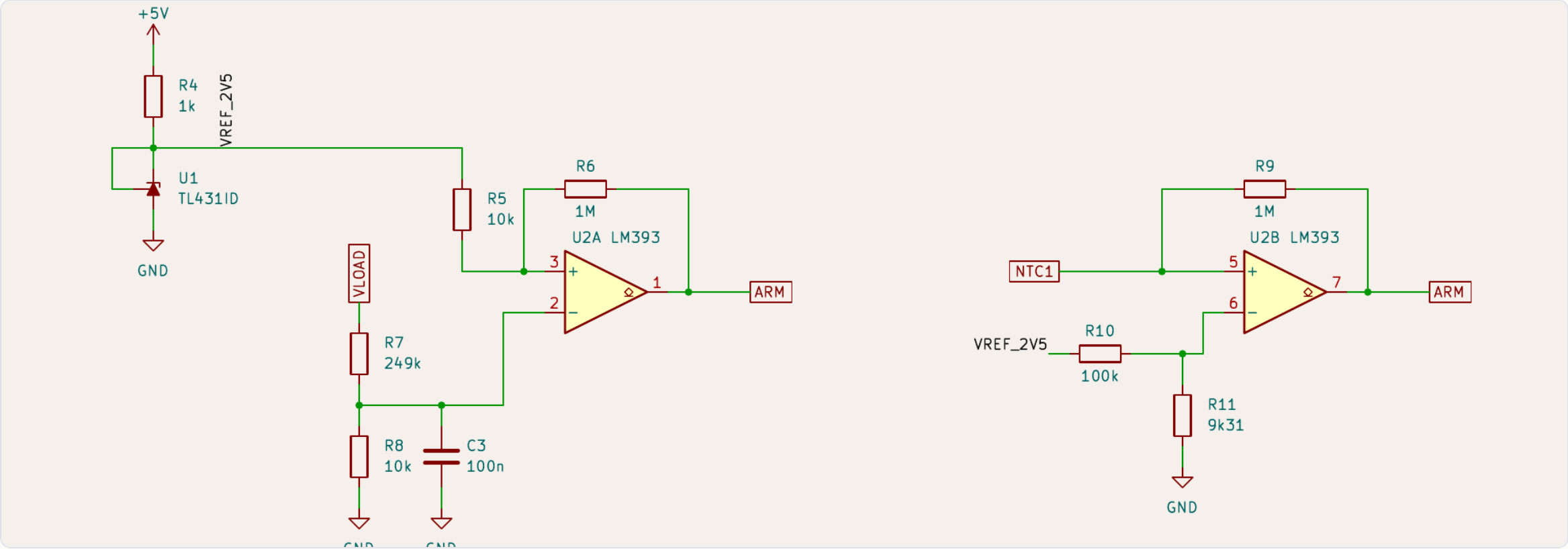
ARM / CLAMP: off unless explicitly enabled



Wired-AND: any source can pull ARM low; only the MCU can lift it, and only if nothing else is pulling it down.

- **ARM is active-high** with a 10 k pull-down. No MCU, MCU in reset, MCU booting, 3.3 V missing → ARM low → load off.
- **CLAMP** does not rely on the DAC: it shorts the setpoints directly, so a DAC glitch at power-up cannot turn a channel on.
- **Boot-safe pins.** ARM_DRV sits on GPIO12, which stays quiet at boot. ARM_FB is on GPIO16 (high at boot) through 100 k, too weak to lift ARM against the 10 k.
- Firmware reads ARM_FB to know when hardware has tripped and reports which source did it.

Hardware over-voltage and over-temperature




```
OVP 2.5 V × (249 k + 10 k) / 10 k = 64.7 V
R6 1 MΩ adds hysteresis, C3 filters spikes
```

```
OTP 2.5 V × 9.31 k / 109.3 k = 0.213 V
NTC1 node at ~100 °C (NTC 10 k B3950)
```

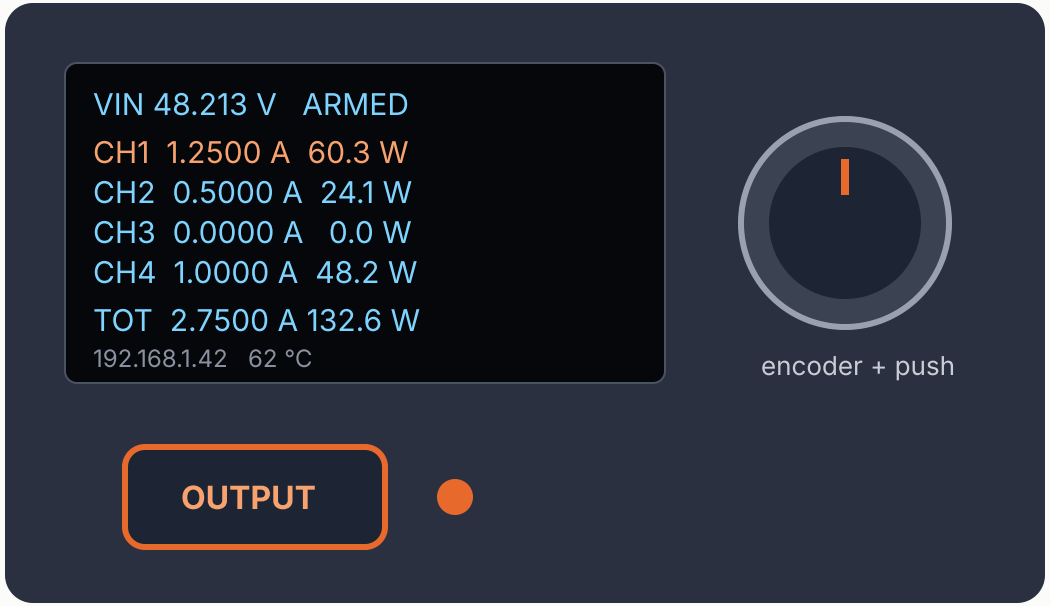
TL431 provides the 2.5 V reference from the analog 5 V. Both LM393 outputs are open collector on ARM, so they work even if the MCU is dead.

Wemos LOLIN D1 mini Pro (ESP8266)

PIN	GPIO	FUNCTION
D1 / D2	5 / 4	I²C SCL / SDA → 4 × INA228
D5 / D7	14 / 13	SPI SCK / MOSI → DAC8565
D8	15	DAC SYNC (low at boot, harmless)
D6	12	ARM_DRV (quiet at boot)
D0	16	ARM_FB read-back (100 k)
D3	0	PCF8574 INT (front panel, rev 0.3)
D4	2	Fan PWM (open-drain via FDS9926A)
RX	3	Fan tach (1 k series)
A0	ADC	NTC1 (on-board 220 k / 100 k divider)

- **Plug-in module** on two 1×8 sockets: no RF layout on the main board; PCB antenna plus a U.FL connector for an external one.
- **Antenna side out:** pin 1 sits at the right board edge and there is no copper beyond x = 164 mm, so the antenna overhangs, away from cooler and spreader.
- **USB is not isolated** — its ground is DUT−. Program with the DUT and fan disconnected; afterwards update over the air.
- firmware re-assign GPIO12 as output after `SPI.begin()`; the HSPI driver claims it as MISO.

Local display and encoder



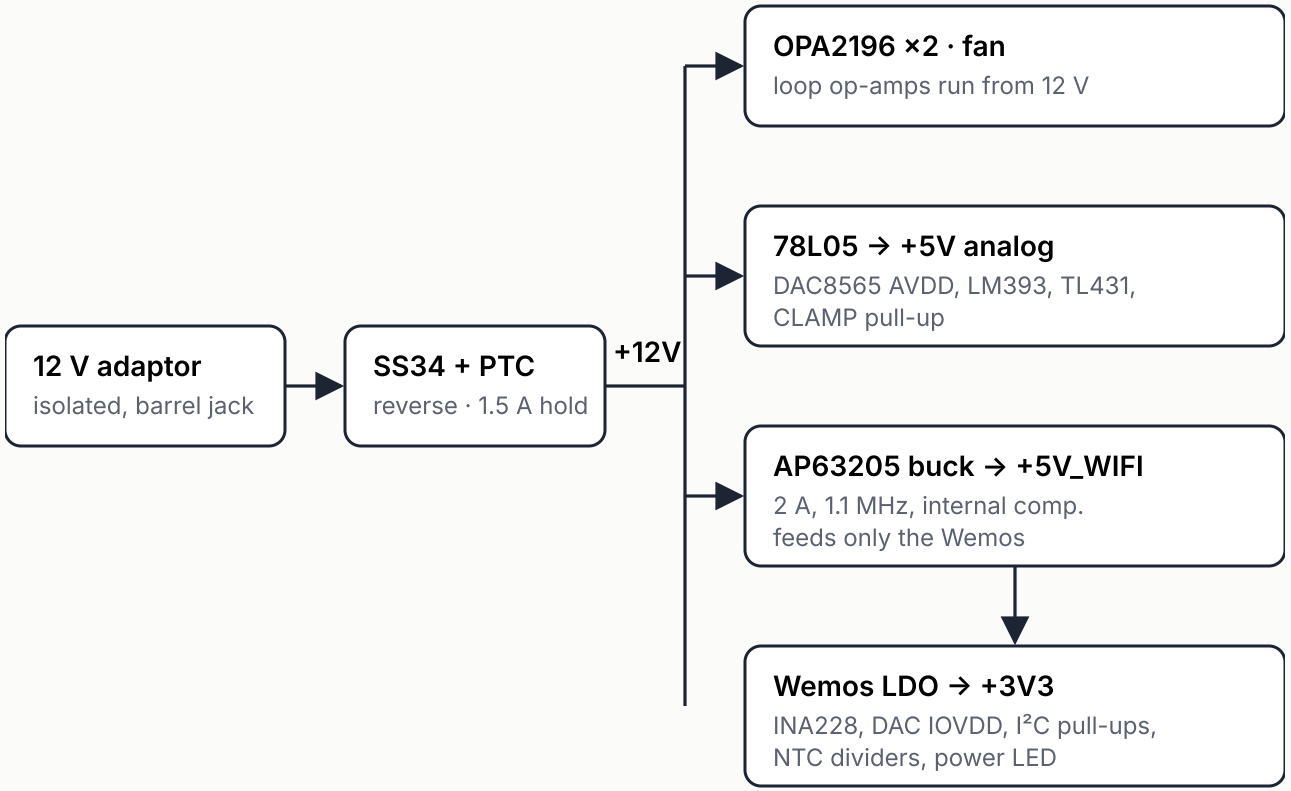
Mock-up of the screen layout (128 × 64 OLED, 2.42″ or 1.3″).

10-pin IDC J6 → front panel

1 3V3 · 2 GND · 3 SDA · 4 GND · 5 SCL
6 ENC_A · 7 ENC_B · 8 ENC_SW · 9 OUTPUT · 10 LED

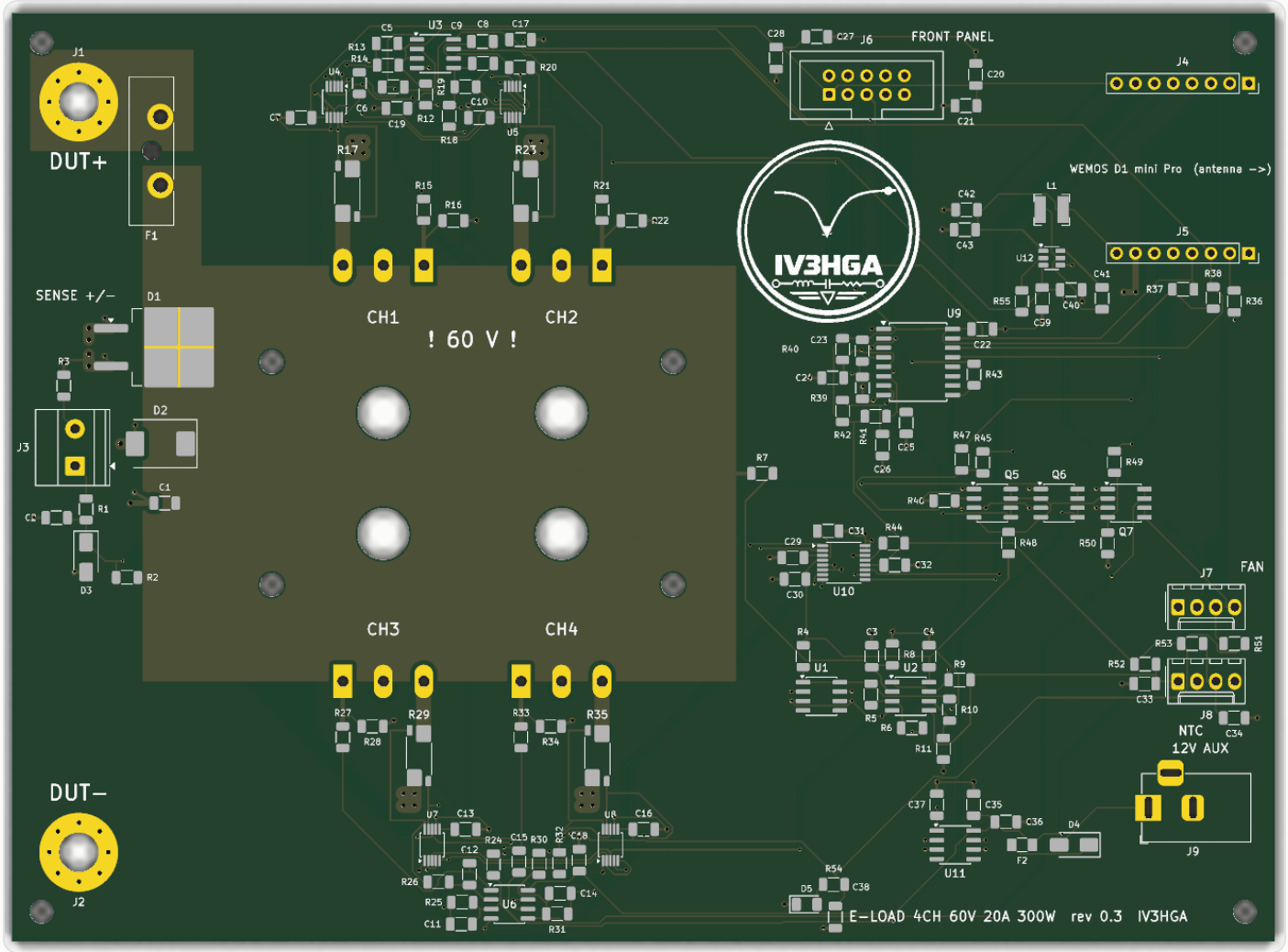
- **OLED on the existing I²C bus** at 0x3C — no conflict with the INA228s (0x40–0x43). One line per channel plus total, bus voltage, ARM state, IP address.
- **PCF8574 I/O expander** (0x20) on the main board reads encoder A/B, push and the OUTPUT button and drives its LED; its INT goes to GPIO0, the only spare pin.
- **Encoder as a vernier:** push toggles coarse / fine step and long-press selects the channel. It is relative, so local and web control never disagree.
- **OUTPUT ON** is the only way to raise ARM from the panel: the load never starts by itself after power-up or a trip.
- **Bus hygiene:** pull-ups lowered to 2.2 k for the ~20 cm ribbon; 10 nF + 10 k debounce (100 µs) on each switch; 10 µF + 100 nF at the connector for the OLED charge pump.
- Turning the encoder during boot can pull GPIO0 low via INT and start the bootloader — harmless, just reset.

Auxiliary power tree



- **Separate 5 V rails.** The WiFi module draws 300–400 mA bursts; the buck isolates those from the DAC's analog supply, which stays on a linear 78L05.
- **Single ground** = DUT−, star point at the J2 stud. The 12 V adaptor must therefore be isolated from mains earth or from the DUT's ground.
- PTC raised to 1.5 A hold to cover buck + fan.

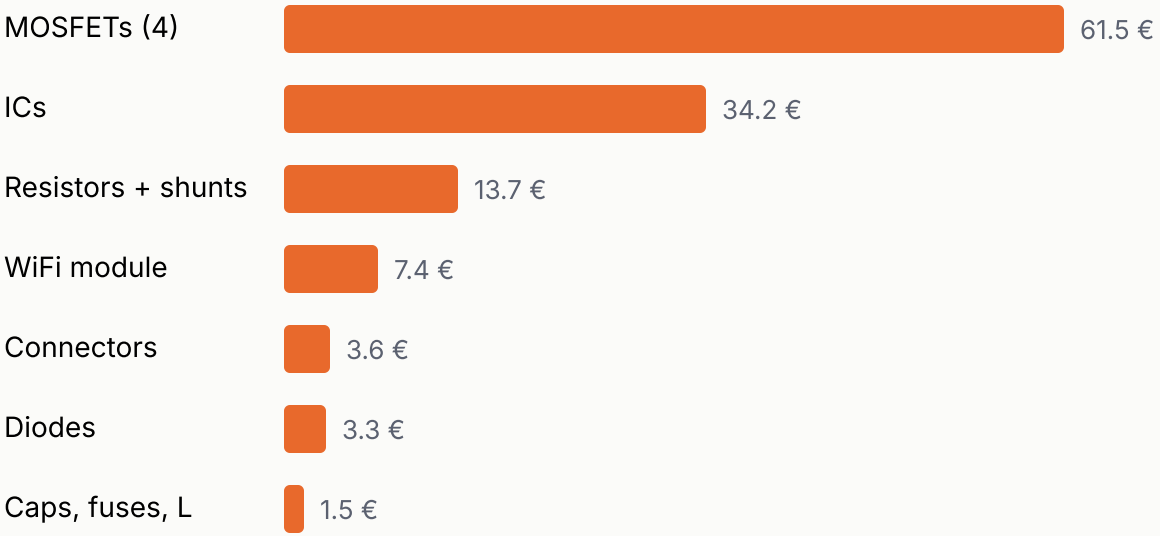
PCB



- **170 × 125 mm**, 2 layers, 2 oz copper, sits on top of the cooler.
- VLOAD as a top-layer pour over the MOSFET field; continuous GND plane on the bottom.
- HV net class: 0.6 mm clearance; local 0.65 mm on HV through-holes.
- Kelvin pairs from each shunt routed by hand; signals auto-routed (Freerouting).
- DRC with schematic parity: 0 unconnected, 0 clearance errors, **0 silkscreen overlaps** — every reference sits next to its part (KiCad 10.0.7).
- Assembly aids: interactive HTML BOM and top/bottom assembly drawings in `docs/assembly`.

COST

Bill of materials



Electronics	**≈ 126 € per board**
Mechanics	≈ 42 € estimated incl. front panel + cooler, spreader, pad (to quote)
PCB	JLCPCB 2 oz — to quote

- Single-quantity distributor prices, VAT excluded, checked October 2026.
- The four linear MOSFETs are half the cost — and the long-lead item (21–38 weeks factory lead time).
- Passive cost is dominated by minimum order quantities, not unit price.
- Full priced BOM with sources: `BOM_prezzi_carico_attivo.xlsx`.